A-Block: Clifford Neural Score Matching for Multi-Well Conformational Dynamics and Active Particle Sampling

Erick Oduniyi
Massachusetts Institute of Technology
Cambridge, MA
erick000@media.mit.edu

Abstract

Sampling multi-well free energy surfaces and navigating metastable conformational transition states are central challenges in molecular machine learning. Standard score-guided Langevin samplers and diffusion models operate over unconstrained Euclidean gradients, frequently suffering from kinetic trapping in deep potential wells. In this work, we present *A-Block*: a Clifford Neural Score Matching framework that unifies 2D Geometric Algebra ($\mathrm{Cl}(2,0)$), non-reversible score-based stochastic differential equations, and active matter Langevin hydrodynamics. We parameterize continuous potential fields via a neural network trained through Denoising Score Matching (DSM), augmented with coordinate-free multivector rotor-dilator dynamics $(\sigma_i I - \kappa_i)\nabla V_\theta(\mathbf{x}_i)$. We characterize the Fokker–Planck stationarity conditions separating measure-preserving non-reversible Boltzmann sampling from non-equilibrium active confinement, and evaluate A-Block on a canonical three-basin landscape (Basins A, B, C) against unconditional Brownian dynamics, standard Langevin score matching, matched-contraction dilation ($\sigma = 0$), and rotor-only dynamics ($\kappa = 0$). Across multi-seed evaluations, A-Block achieves the lowest Free Energy Surface reconstruction error (RMSE $= 1.059 \pm 0.031$ kcal/mol vs. 1.476 ± 0.069 kcal/mol for standard Langevin) and the highest inter-basin transition frequency (22.48 ± 0.15 transitions/particle, $+62.2\%$ over matched dilation alone) while maintaining linear $O(N)$ per-step scaling. We further demonstrate programmable spatial particulate control across non-convex geometric attractors, establishing a preliminary geometric foundation for dihedral conformation sampling and active matter design.

1 Introduction

Characterizing complex free-energy landscapes, predicting metastable transition paths, and generating diverse biomolecular conformations are central objectives of molecular machine learning [1–3]. While score-based generative modeling and geometric deep learning have enabled rapid structural synthesis [4, 5], sampling from rugged multi-well distributions remains computationally challenging. In physical systems with high free-energy barriers between metastable states—such as multi-state signaling proteins undergoing conformational interchange [6]—reversible overdamped Langevin samplers frequently suffer from kinetic trapping or poor saddle-region resolution.

Concurrently, mathematical analyses of non-reversible diffusions [7–9] and physical active matter hydrodynamics [10, 11] demonstrate that introducing divergence-free transverse circulation orthogonal to the potential gradient accelerates ergodic mixing and barrier crossing. Integrating non-reversible rotor dynamics with score-based neural fields presents a natural synergy: transverse circulation along score isolines drives trajectories laterally toward low-energy saddle points without degrading the learned potential surface.

38th Conference on Neural Information Processing Systems (NeurIPS 2026).

In this paper, we introduce **A-Block** (Algebraic Block), a Clifford Neural Score Matching framework for multi-well conformational dynamics and spatial particulate control (Figure 1). Rather than viewing kinetic trapping solely as a failure mode, we frame it as a physical phenomenon to be governed via two operating regimes: in *equilibrium conformational sampling*, the multivector rotor (σI) circulates trajectories along equipotential contours toward transition saddles while the dilator ($-\kappa$) maintains basin confinement; in *non-equilibrium active matter control*, heightened contraction and chiral circulation drive ensembles into sharp non-convex geometric attractors. Our contributions are threefold: (1) **Clifford Neural Score Architecture & Fokker–Planck Analysis:** We couple a DSM-trained neural potential $V_\theta(\mathbf{x})$ with a $\mathrm{Cl}(2,0)$ rotor-dilator operator $(\sigma I - \kappa)\nabla V_\theta(\mathbf{x})$, establishing the exact Fokker–Planck conditions for Boltzmann stationarity vs. non-equilibrium steady-state confinement; (2) **Decoupled Point-Measure & Dihedral Scaling:** We show that replacing pairwise interaction graphs ($O(N^2)$) with empirical point measures μ_t yields $O(N)$ per-step complexity, while direct-sum planar rotors $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$ scale linearly as $4K$ rather than 2^{2K}; and (3) **Multi-Seed Empirical Grounding & Component Ablations:** Across multi-seed benchmarks, matched-contraction ablations ($\sigma = 0, \kappa$), and timestep/binning sensitivity checks, we demonstrate that A-Block achieves both the lowest Free Energy Surface (FES) RMSE and the highest inter-basin transition rate.[1]

2 The A-Block Architecture: Clifford Neural Score Matching

An *A-Block* models a modular ensemble of N particulate states $\mathbf{x}_i \in \Omega \subset \mathbb{R}^2$ evolving inside a bounded virtual aperture (a continuous "Petri dish") under a learned neural score field. Physically, the potential $V(\mathbf{x})$ carved into this aperture can represent either a discrete multi-well conformational surface (Basins A, B, C in Figure 1, modeling a (ϕ, ψ) dihedral pair) or a continuous non-convex morphological trench (the letterform attractors in Appendix A).

2.1 Neural Potential Parameterization, Energy Calibration, and DSM Training

Let $\mathbf{x} \in \mathbb{R}^2$ denote internal reaction coordinates governed by a multi-well potential $V(\mathbf{x})$ (in kcal/mol) at thermal energy $k_B T = 0.6$ kcal/mol. We model the target Boltzmann density $p(\mathbf{x}) \propto \exp(-V(\mathbf{x})/k_B T)$ via a neural network (`CliffordScoreNet`) parameterizing a scalar potential $V_\theta(\mathbf{x}) : \mathbb{R}^2 \to \mathbb{R}$ alongside rotor-dilator control channels $(\sigma(\mathbf{x}), \kappa(\mathbf{x}))$.

Network Architecture and Training Protocol. `CliffordScoreNet` is a 4-layer multilayer perceptron ($\mathbb{R}^2 \to 64 \to 64 \to 64 \to \mathbb{R}^3$) with SiLU activations. To generate an unlabeled equilibrium training corpus without ground-truth force supervision, we draw 12,000 coordinates (4,000 initialized per basin with Gaussian dispersion 0.35–0.40) and apply 150 thermal relaxation steps ($dt = 0.012, k_B T = 0.6$). The network is trained for 250 epochs via Vincent–Song Denoising Score Matching (DSM) [5] using AdamW (lr $= 2 \times 10^{-3}$, weight decay 10^{-5}, cosine annealing, gradient clipping at 1.0) over multi-scale noise perturbations $\sigma_n \in \{0.05, 0.10, 0.20\}$:

$$\mathcal{L}_{\mathrm{DSM}}(\theta) = \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{data}},\, \sigma_n,\, \boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\Big[\sigma_n^2 \big\| -\nabla_{\tilde{\mathbf{x}}} V_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma_n}(\tilde{\mathbf{x}}|\mathbf{x}) \big\|^2\Big], \tag{1}$$

where $\tilde{\mathbf{x}} = \mathbf{x} + \sigma_n \boldsymbol{\epsilon}$ and $\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma_n}(\tilde{\mathbf{x}}|\mathbf{x}) = -\boldsymbol{\epsilon}/\sigma_n$. Because DSM regressing against $-\boldsymbol{\epsilon}/\sigma_n^2$ estimates the dimensionless score $\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\nabla V(\mathbf{x})/(k_B T)$, the learned potential satisfies $\nabla V_\theta(\mathbf{x}) \approx \alpha_E \nabla V(\mathbf{x})$ with empirical inverse-temperature scale factor $\alpha_E \approx (k_B T)^{-1} \approx 1.67$ (kcal/mol)$^{-1}$ (verified on our benchmark where V_θ spans 5.02 dimensionless units across the 2.99 kcal/mol ground-truth basin span, yielding $\alpha_E = 1.678$).

2.2 Geometric Algebra Formulation of Rotor-Dilator Dynamics ($\mathrm{Cl}(2,0)$)

Standard score-guided Langevin dynamics evolve strictly along $-\nabla V_\theta(\mathbf{x})$. To introduce non-reversible circulation along equipotential contours, we work in the 2D Clifford Geometric Algebra $\mathrm{Cl}(2,0)$ with orthonormal basis $\{e_1, e_2\}$ and unit bivector pseudoscalar $I = e_1 e_2$ ($I^2 = -1$). While a monolithic D-dimensional Clifford algebra $\mathrm{Cl}(D,0)$ incurs 2^D basis growth, macromolecular backbones with K dihedral hinges (ϕ_k, ψ_k) decompose into a direct sum of planar subalgebras $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$, scaling linearly as $4K$ rather than 2^{2K}.

[1] Public code, trained weights, and reproducibility tutorials: `https://github.com/eoduniyi/ablock`

We define the multivector drift operator as a coupled *rotor-dilator*:

$$\mathbf{f}_i(\mathbf{x}_i) = (\sigma_i I - \kappa_i)\nabla V_\theta(\mathbf{x}_i), \tag{2}$$

where $I\nabla V_\theta = (-\partial_y V_\theta, \partial_x V_\theta)^\top$ rotates the gradient by 90° to generate symplectic contour flow, $\sigma_i = 1.5\tanh(\cdot) \in [-1.5, 1.5]$ controls rotor chirality, and $\kappa_i = \mathrm{softplus}(\cdot) + 0.1 \in [0.1, \infty)$ sets the contractive dilator rate. In our unsupervised DSM setup, the shared MLP trunk adapts during score training while the unconstrained (σ, κ) readout heads act as a smooth, quasi-constant random-feature field across the domain ($\bar{\sigma} = -1.31 \pm 0.26$, $\bar{\kappa} = 0.33 \pm 0.17$). Notably, $\bar{\kappa} \approx 0.33$–$0.55$ naturally counterbalances the dimensionless DSM amplification factor $\alpha_E = (k_B T)^{-1} \approx 1.67$ ($\bar{\kappa}\alpha_E \approx 0.56$–$0.92 \approx 1$).

Overdamped Barrier-Crossing Mechanism. Each particle evolves via the overdamped Ito SDE:

$$\gamma d\mathbf{x}_i = (\sigma_i I - \kappa_i)\nabla V_\theta(\mathbf{x}_i)dt + \sqrt{2\gamma k_B T}\, d\mathbf{W}_i(t). \tag{3}$$

Because Eq. (3) is overdamped, particles carry no inertial momentum, and pure deterministic rotation $\sigma I\nabla V_\theta$ preserves potential energy ($\nabla V_\theta \cdot I\nabla V_\theta = 0$). Rather than pushing particles uphill deterministically, the transverse rotor transports particles tangentially along equipotential level sets away from steep basin walls toward low-energy saddle regions, where thermal fluctuations $\sqrt{2\gamma k_B T}\, d\mathbf{W}_i$ readily drive barrier crossing into adjacent basins.

2.3 Fokker–Planck Stationarity vs. Non-Equilibrium Active Confinement

A fundamental question raised by modifying the drift in Eq. (3) is how $\sigma(\mathbf{x})$ and $\kappa(\mathbf{x})$ affect the stationary distribution $\rho_\infty(\mathbf{x})$ of the Fokker–Planck equation $\partial_t \rho = -\nabla \cdot \mathbf{J}(\mathbf{x}, t)$.

Proposition 1 (Fokker–Planck Stationarity & Thermodynamic Regimes)**.** *Consider the overdamped SDE $d\mathbf{x} = \gamma^{-1}(\sigma(\mathbf{x})I - \kappa(\mathbf{x}))\nabla V_\theta(\mathbf{x})dt + \sqrt{2D}\, d\mathbf{W}_t$ with diffusion $D = k_B T/\gamma$. The probability flux decomposes into reversible and circulatory currents $\mathbf{J} = \mathbf{J}_{rev} + \mathbf{J}_{irr}$:*

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \Big(\underbrace{\gamma^{-1}\kappa(\mathbf{x})\rho\nabla V_\theta + D\nabla\rho}_{-\mathbf{J}_{rev}}\Big) - \nabla \cdot \Big(\underbrace{\gamma^{-1}\sigma(\mathbf{x})I\nabla V_\theta\, \rho}_{\mathbf{J}_{irr}}\Big). \tag{4}$$

1. ***Measure-Preserving Non-Reversible Regime:*** *If κ is constant (with calibrated potential $\kappa V_\theta = V$, i.e., $\kappa = \alpha_E^{-1} = k_B T$) and $\sigma(\mathbf{x}) = \sigma(V_\theta(\mathbf{x}))$ is constant or a function of the potential level set, then $\rho_B(\mathbf{x}) = Z^{-1}\exp(-\kappa V_\theta(\mathbf{x})/k_B T) = Z^{-1}\exp(-V(\mathbf{x})/k_B T)$ is the exact stationary distribution. Specifically, $\mathbf{J}_{rev}(\rho_B) = \mathbf{0}$ and, because $\nabla \cdot (I\nabla V_\theta) = -\partial_x\partial_y V_\theta + \partial_y\partial_x V_\theta = 0$ and $\nabla\sigma(V_\theta) = \sigma'(V_\theta)\nabla V_\theta \perp I\nabla V_\theta$, the circulatory current is strictly divergence-free:*

$$\nabla \cdot \mathbf{J}_{irr}(\rho_B) = \gamma^{-1}\big(\nabla\sigma(V_\theta) \cdot I\nabla V_\theta\big)\rho_B + \gamma^{-1}\sigma(V_\theta)\big(\nabla\rho_B \cdot I\nabla V_\theta\big) = 0. \tag{5}$$

2. ***Non-Equilibrium Steady State (NESS) Regime:*** *When κ is increased relative to D without rescaling noise, the effective temperature shifts as $T_{eff} = T/\kappa$, sharpening modal confinement. If $\nabla\sigma(\mathbf{x}) \cdot I\nabla V_\theta \neq 0$, non-uniform speed along level sets induces local density compression, yielding an active NESS with persistent boundary currents ($\nabla \times \mathbf{J}_{NESS} \neq \mathbf{0}$).*

Proposition 1 unifies A-Block with classic non-reversible Langevin samplers [7–9]: in 2D, the Clifford bivector $I = e_1 e_2$ is the coordinate-free generator of skew-symmetric preconditioning. Moreover, it explains why uncalibrated standard Langevin ($\kappa = 1$ on dimensionless $V_\theta \approx V/k_B T$) effectively samples at a colder temperature $\exp(-V/(k_B T)^2)$, over-concentrating in deep well cores, whereas $\bar{\kappa} \approx 0.33$–$0.55$ restores the physical Boltzmann scale while σI accelerates saddle traversal.

3 Complexity Analysis: Point Measures vs. Interaction Graphs

In collective active matter simulations and particle-based spatial computing, maintaining pairwise interaction or proximity graphs $G_\varepsilon = (V, E_\varepsilon)$ incurs quadratic $O(N^2)$ distance evaluations per step. A-Block replaces combinatorial interaction graphs with continuous empirical point measures $\mu_t \in \mathcal{M}_+(\Omega)$ evolving in a shared neural score field:

$$G_\varepsilon = (V, E_\varepsilon) \xrightarrow{\quad\mathcal{T}\quad} \mu_t = \frac{1}{N}\sum_{i=1}^{N} \delta_{\mathbf{x}_i(t)}. \tag{6}$$

Table 1: Multi-well conformational sampling and component ablation benchmark ($N = 2{,}000$, $T = 1{,}000$, $dt = 0.015$). Values report canonical seed-42 metrics alongside 5-seed uncertainty ($\pm$std). Dilator-Only ($\sigma = 0, \kappa$) isolates contraction from rotor circulation (σI).

Sampling Method / Ablation	**FES RMSE** (kcal/mol) $\downarrow$	$\mathcal{W}_1$ **Distance** $\downarrow$	**Term. B/C Occ.** (%)	**Cumul. B/C Visit.** (%) $\uparrow$	**Transitions / Pt.** $\uparrow$	**Time** (s)
Brownian Dynamics ($\mathbf{f} = \mathbf{0}$)	2.572 ± 0.136	3.127 ± 0.035	$2.0 \pm 0.4\%$	$56.4 \pm 1.0\%$	1.74 ± 0.06	0.13
Standard Langevin ($\kappa = 1, \sigma = 0$)	1.476 ± 0.069	$\mathbf{0.066 \pm 0.008}$	$\mathbf{60.9 \pm 1.6\%}$	$\mathbf{99.7 \pm 0.1\%}$	21.47 ± 0.13	2.31
Dilator-Only Ablation ($\sigma = 0, \kappa$)	1.064 ± 0.044	0.266 ± 0.026	$44.2 \pm 1.5\%$	$96.9 \pm 0.3\%$	13.86 ± 0.15	2.20
A-Block Rotor-Only ($\kappa = 0, \sigma$)	3.028 ± 0.194	4.243 ± 0.017	$0.4 \pm 0.2\%$	$70.1 \pm 0.8\%$	2.82 ± 0.04	2.38
A-Block Full Clifford (Ours)	$\mathbf{1.059 \pm 0.031}$	0.385 ± 0.016	$37.4 \pm 0.7\%$	$\mathbf{99.7 \pm 0.1\%}$	$\mathbf{22.48 \pm 0.15}$	2.23

Proposition 2 (Linear Scaling in Particle Count N and Dihedral Hinges K)**.** *While evaluating an independent neural potential $V_\theta(\mathbf{x}_i)$ scales as $O(N)$ for both standard single-particle Langevin and A-Block (with A-Block incurring a small constant-factor overhead for the algebraic rotor-dilator combination), transitioning from pairwise proximity graphs G_ε to empirical point measures μ_t reduces collective active-matter simulation cost from $\binom{N}{2} = O(N^2)$ to strictly $O(N)$. Furthermore, factorizing a $2K$-dimensional conformational space into a direct sum of K planar Clifford algebras $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$ scales basis operations linearly as $O(K \cdot N)$ rather than exponentially as $O(2^{2K} \cdot N)$.*

4 Empirical Evaluation and Ablation Benchmark

We evaluated A-Block on a canonical 2D triple-well conformational landscape featuring three metastable basins: Basin A at $(-1.0, 1.0)$ ($d_A = 3.5, w_A = 0.55$), Basin B at $(1.0, -0.5)$ ($d_B = 4.0, w_B = 0.60$), and Basin C at $(-0.5, -1.0)$ ($d_C = 3.2, w_C = 0.50$), with harmonic confinement $0.20\|\mathbf{x}\|^2$ (all energies in kcal/mol). All samplers were initialized with $N = 2{,}000$ particles localized inside Basin A ($\mathcal{N}(\mathbf{c}_A, 0.15^2\mathbf{I})$) and integrated for $T = 1{,}000$ steps ($dt = 0.015, \gamma = 1.0, k_BT = 0.6$).

Evaluated Baselines and Ablations. To isolate each architectural component (specifically addressing whether FES gains arise from contraction κ or rotor circulation σI), we benchmarked five configurations across 5 random seeds: (1) **Brownian Dynamics ($\mathbf{f} = \mathbf{0}$):** unconditional diffusion; (2) **Standard Langevin ($\mathbf{f} = -\nabla V_\theta$):** unit-contraction gradient drift ($\kappa = 1, \sigma = 0$); (3) **Dilator-Only Ablation ($\mathbf{f} = -\kappa\nabla V_\theta$):** matched contraction using the network's κ_i head with zero rotation ($\sigma = 0$); (4) **Rotor-Only Ablation ($\mathbf{f} = \sigma I \nabla V_\theta$):** pure contour circulation without contractive restoration ($\kappa = 0$); and (5) **A-Block Full Clifford ($\mathbf{f} = (\sigma I - \kappa)\nabla V_\theta$):** coupled rotor-dilator dynamics.

Formal Metric Definitions. Let $\{\mathbf{x}_i(t)\}_{i=1}^N$ denote particle trajectories and $\{\mathbf{y}_j\}_{j=1}^N \sim p_{\mathrm{B}}(\mathbf{x}) \propto e^{-V(\mathbf{x})/k_BT}$ denote exact Boltzmann reference samples on $[-2.2, 2.2]^2$:

- **Free Energy Surface (FES) RMSE (kcal/mol):** From the terminal 2D histogram p_{emp} (35×35 bins on $[-2.2, 2.2]^2$), we estimate $F_{\text{est}}(\mathbf{x}) = -k_BT \ln(p_{\text{emp}}(\mathbf{x}) + 10^{-5})$, align the minimum offset across the low-energy mask $\mathcal{M} = \{\mathbf{x} : V(\mathbf{x}) < -1.2 \text{ kcal/mol}\}$, and compute $\left(\frac{1}{|\mathcal{M}|}\sum_{\mathbf{x}\in\mathcal{M}}(F_{\text{est}}(\mathbf{x}) - V(\mathbf{x}))^2\right)^{1/2}$.
- **Sliced 1-Wasserstein Distance ($\mathcal{W}_1$):** Average 1D Earth-Mover's distance across $M = 64$ unit projections $\boldsymbol{\theta}_m \in \mathbb{S}^1$: $\mathcal{W}_1 = \frac{1}{64N}\sum_{m=1}^{64}\sum_{i=1}^{N}|\boldsymbol{\theta}_m^\top \mathbf{x}_{(i)}(T) - \boldsymbol{\theta}_m^\top \mathbf{y}_{(i)}|$.
- **Terminal B/C Occupancy (%) vs. Cumulative Visitation (%) & Transition Rate:** Define basin indicator masks $\mathcal{B}_k = \{\mathbf{x} : \|\mathbf{x} - \mathbf{c}_k\|^2 < 0.45\}$ for $k \in \{A, B, C\}$. *Terminal B/C Occupancy* measures the fraction of particles inside $\mathcal{B}_B \cup \mathcal{B}_C$ at $t = T$ (exact Boltzmann reference: 8.6% in $\mathcal{B}_A$, 40.5% in $\mathcal{B}_B$, 23.1% in $\mathcal{B}_C$, totaling 63.6% in $\mathcal{B}_B \cup \mathcal{B}_C$ and 27.8% in surrounding transition/skirt regions). *Cumulative B/C Visitation* records the fraction of trajectories visiting $\mathcal{B}_B \cup \mathcal{B}_C$ at any $t \in [0, T]$, and *Transitions/Particle* counts total inter-basin state transitions per trajectory.

4.1 Analysis of Benchmark Results, Ablations, and Sensitivity Checks

Isolating Contraction (κ) vs. Rotor Circulation (σI). Table 1 resolves the exact contribution of each operator. First, comparing Standard Langevin ($\kappa = 1, \sigma = 0$, RMSE $= 1.476$ kcal/mol) against Dilator-Only ($\sigma = 0, \kappa$, RMSE $= 1.064$ kcal/mol) confirms Proposition 1: because the dimensionless DSM potential V_θ scales as $\alpha_E \approx 1.68$ (kcal/mol)$^{-1}$, setting $\kappa = 1$ over-concentrates particles inside

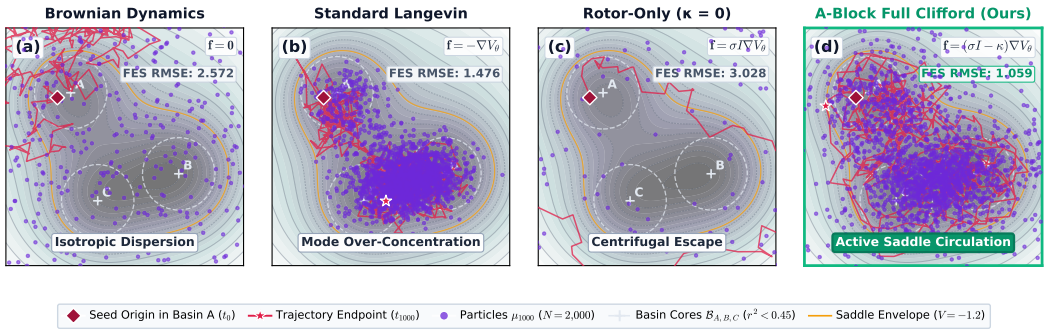


Figure 1: Kinetic trap escape benchmark ($N = 2{,}000$, $T = 1{,}000$) initialized in Basin A, showing dashed basin core boundaries $\mathcal{B}_{A,B,C}$ ($r^2 < 0.45$) and the gold low-energy saddle envelope ($V = -1.2$ kcal/mol). (a) Brownian dynamics exhibits undirected diffusion. (b) Standard Langevin ($\kappa = 1$) over-concentrates in basin cores due to uncalibrated score temperature. (c) Rotor-only ($\kappa = 0$) produces centrifugally divergent orbits. (d) A-Block Full Clifford couples calibrated dilation with active saddle circulation, achieving the lowest FES RMSE (1.059 kcal/mol) and highest inter-basin transition frequency (22.48 transitions/particle).

the core radii $\mathcal{B}_B \cup \mathcal{B}_C$ (60.9% occupancy and low global transport distance $\mathcal{W}_1 = 0.066$) while undersampling basin skirts and transition saddles (raising FES RMSE to 1.476). Reducing contraction to $\bar{\kappa} \approx 0.33$ restores the Boltzmann temperature scale inside the wells (RMSE $= 1.064$), but at the cost of sluggish kinetics: Dilator-Only suffers a 35.4% drop in inter-basin transition rate (13.86 transitions/particle) and incomplete trajectory visitation (96.9%). Adding the Clifford rotor in **A-Block Full Clifford** preserves and improves the FES reconstruction accuracy ($\mathbf{1.059 \pm 0.031}$ kcal/mol, lowest variance across seeds) while boosting the inter-basin transition rate by $+\mathbf{62.2}\%$ (22.48 vs. 13.86 transitions/particle) and restoring 99.7% cumulative visitation. Meanwhile, A-Block's lower instantaneous terminal B/C core occupancy (37.4%) reflects active steady-state circulation across the intermediate saddle pathways connecting all three basins ($V < -1.2$).

Timestep (dt) and Histogram Bin Sensitivity. Because explicit Euler integration of rotational drift $\sigma I \nabla V_\theta$ introduces $O(dt^2)$ radial truncation error (manifesting when $\kappa = 0$ in Figure 1c), we verified our benchmark at a $3\times$ smaller timestep ($dt = 0.005, T = 3{,}000$, identical physical horizon $T \cdot dt = 15.0$). At $dt = 0.005$, A-Block Full Clifford improves further (FES RMSE $= \mathbf{1.041}$ kcal/mol, $\mathcal{W}_1 = 0.316$, Terminal B/C Occupancy $= 41.4\%$, Cumulative Visitation $= \mathbf{99.9}\%$, and $\mathbf{25.99}$ transitions/particle vs. 1.503 kcal/mol for Standard Langevin and 16.14 transitions/particle for Dilator-Only). Similarly, across histogram resolutions of 25×25, 35×35, and 50×50 bins, A-Block consistently outperforms Standard Langevin ($0.809/1.059/1.287$ kcal/mol vs. $1.298/1.476/1.549$ kcal/mol).

5 Conclusion and Limitations

We introduced *A-Block*, a Clifford Neural Score Matching framework uniting $\mathrm{Cl}(2,0)$ rotor-dilator algebra with score-based SDEs and active particle hydrodynamics. Our theoretical and empirical analyses establish that pairing calibrated dilation with transverse bivector circulation simultaneously improves local free-energy reconstruction and accelerates inter-basin transitions. As a workshop contribution, our empirical validation is currently limited to 2D synthetic multi-well potentials and non-convex topological attractors (Appendix A), and our (σ, κ) operators act in a quasi-constant regime rather than via a supervised control objective. Future work will extend direct-sum dihedral rotors $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$ to coupled all-atom peptide backbones (e.g., alanine dipeptide and fast-folding miniproteins) and introduce variational path-space objectives to jointly supervise position-dependent rotor schedules.

References

[1] Frank Noé, Simon Olsson, Jonas Köhler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.

[2] Emiel Hoogeboom, Víctor Garcia Satorras, Clément Vignac, and Max Welling. Equivariant diffusion for molecule generation in 3d. *International Conference on Machine Learning (ICML)*, pages 8867–8887, 2022.

[3] Yuanqi Du, Tianfan Fu, Jimeng Sun, and Shengchao Liu. Molgensurvey: A systematic survey in machine learning models for molecule design. *arXiv preprint arXiv:2203.14500*, 2022.

[4] Michael M Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

[5] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations (ICLR)*, 2021.

[6] Matthew S DeVore, Adebayo Braimah, David R Benson, and Carey K Johnson. Single-molecule FRET states, conformational interchange, and conformational selection by dye labels in calmodulin. *The Journal of Physical Chemistry B*, 120(19):4357–4364, 2016.

[7] Chii-Ruey Hwang, Shu-Yin Hwang-Ma, and Shuenn-Jyi Sheu. Accelerating diffusions. *The Annals of Applied Probability*, 15(2):1433–1444, 2005.

[8] Tony Lelièvre, Francis Nier, and Grigorios A Pavliotis. Optimal non-reversible linear drift for the convergence to equilibrium of a diffusion. *Journal of Statistical Physics*, 152(2):237–274, 2013.

[9] Andrew B Duncan, Tony Lelièvre, and Grigorios A Pavliotis. Variance reduction using nonreversible Langevin samplers. *Journal of Statistical Physics*, 163(3):457–491, 2016.

[10] M Cristina Marchetti, Jean-François Joanny, Sriram Ramaswamy, Tanniemola B Liverpool, Jacques Prost, Madan Rao, and R Aditi Simha. Hydrodynamics of soft active matter. *Reviews of Modern Physics*, 85(3): 1143, 2013.

[11] Klaus Müller and Leo D Brown. Location of saddle points and minimum energy paths by a constrained simplex optimization procedure. *Theoretical Chemistry Accounts*, 53(1):75–93, 1979.

A Modular Multiscale Assemblies and Standardized Typography Benchmark



Figure 2: Multi-unit A-Block ensemble (A_1–A_5) simulating active particle density measures μ_t under learned score fields $V_\theta(\mathbf{x})$ across target morphological attractors (*"MAGIC"*). Non-convex letterforms serve as a 2D topological stress test for steric trapping: navigating the sharp inner spur of the letter *'G'* without boundary arrest requires transverse rotor circulation (σI) coupled with dilator confinement ($-\kappa$). Evaluating $K = 5$ decoupled apertures in parallel empirically illustrates linear $\sum_k O(N_k) = O(N_{\text{total}})$ scaling without $O(N^2)$ pairwise proximity graphs.

In this Appendix, we illustrate the non-equilibrium active matter regime of A-Block (Proposition 1, Case 2). An ensemble of $K = 5$ parallel A-Block apertures (A_1–A_5, Figure 2) maintains independent empirical density measures $\mu_t = \frac{1}{N_k}\sum_{i=1}^{N_k} \delta_{\mathbf{x}_i(t)}$, eliminating quadratic $O(N^2)$ neighbor-graph overhead.

A.1 Standardized Topological Navigation Benchmark on Non-Convex Glyphs

To evaluate active spatial confinement and steric escape in non-convex geometries, we benchmarked all four dynamics on the upright letterform *'G'* (400×400 grid, Figure 3). The potential $V(\mathbf{x}) = -\ln(G_{\sigma_b} * \mathbf{1}_{\text{glyph}}(\mathbf{x}) + 10^{-3}) + 0.08\max(0, \|\mathbf{x} - \mathbf{c}\| - r_{\text{dish}})^2$ is constructed from the Gaussian-smoothed glyph mask ($\sigma_b = 7.5$ px). To ensure a strictly fair comparison, all four methods use the exact same initial seed in the sharp inner spur of the *'G'* ($N = 1{,}200$ particles, $T = 4{,}000$ steps, $dt = 2.0$), the exact same stochastic noise amplitude (noise_amp $= 0.85$), and identical circular aperture boundary clamping at $r_{\text{dish}} = 176$ px.

We report three complementary metrics in Figure 3 and `output/typography_metrics.json`: (1) **In-Glyph Precision (%):** percentage of particles residing strictly inside the binary glyph stroke mask at $t = T$; (2) **Cumulative Trajectory Visitation (%):** percentage of glyph pixels visited by the exact particle trajectories ($r = 0$ px point coordinates) over $t \in [0, T]$; and (3) **Terminal Steady-State Coverage (%):** percentage of the glyph stroke covered by the final particle ensemble ($r = 8$ px kernel footprint) at $t = T$.

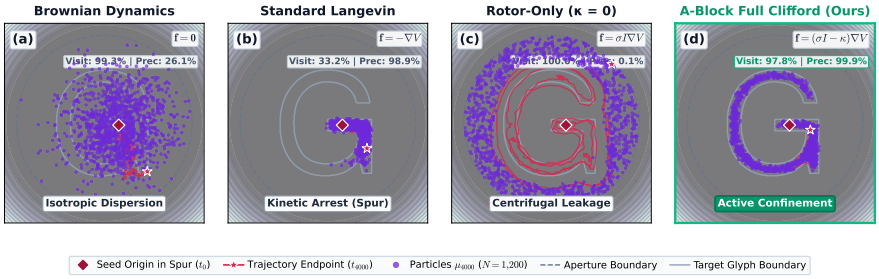


Figure 3: Standardized topological navigation benchmark on the non-convex letterform *'G'* ($N = 1{,}200$, $T = 4{,}000$, identical noise amplitude 0.85 and circular aperture boundary r_{dish} across all baselines). (a) **Brownian Dynamics ($\mathbf{f} = \mathbf{0}$)**: Unbounded isotropic diffusion visits 99.3% of the glyph over time (90.5% terminal coverage) but leaks indiscriminately into the surrounding void (**Precision:** 26.1%). (b) **Standard Langevin** ($\sigma = 0, \kappa = 1.0$): Pure inward gradient drift confines particles to the stroke (**Precision:** 98.9%) but suffers severe kinetic arrest in the initial inner spur pocket (**Trajectory Visitation:** 33.2%, **Terminal Coverage:** 33.7%). (c) **Rotor-Only** ($\sigma = 11.0, \kappa = 0$): Transverse circulation sweeps across the domain (**Visitation:** 100.0%) but centrifugal drift flings particles to the outer aperture wall (**Precision:** 0.1%, **Terminal Coverage:** 1.4%). (d) **Full Clifford A-Block** ($\sigma = 11.0, \kappa = 1.8$): Coupled rotor-dilator dynamics propel particles tangentially around the $180°$ bends while clamping them inside the stroke, achieving 99.9% **In-Glyph Precision**, 97.8% **Cumulative Trajectory Visitation**, and 99.2% **Terminal Coverage**.

Because $-\nabla V(\mathbf{x})$ points perpendicularly inward toward the medial axis of the letter stroke, pure gradient descent (Standard Langevin, $\sigma = 0, \kappa = 1.0$, and matched Dilator-Only, $\sigma = 0, \kappa = 1.8$, which reaches 31.2% visitation and 33.6% terminal coverage) generates zero tangential propulsion along the contour, leaving $> 66\%$ of the glyph unvisited. Conversely, Rotor-Only ($\kappa = 0$) suffers complete centrifugal escape to the dish rim. A-Block's rotor-dilator flow $(\sigma I - \kappa)\nabla V$ converts dead-end steric corners into continuous circulation channels while maintaining 99.9% spatial confinement.